MANYARA REGION

FORM FOUR MOCK EXAMINATION 2021

032/1 **CHEMISTRY 1**

(For both school and private candidates)

MARKING SCHEME

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	E	C	A	A	B	C	A	A	A

(10 marks@ 01 mark)

2.

i	ii	iii	iv	v
F	C	E	A	D

(05 marks @ 01 mark)

3. (a)(i) A good fuel should burn with a moderate velocity**. (01 Mark)**
(ii) It should have an average ignition point. **(01 Mark)**
(iii) It should not give off poisonous gases during combustion. **(01 Mark)**
(iv) Should have light parametric effects. **(01 Mark)**
(v) Should be easy and safe to transport. **(01 Mark)**
(vi) Should be affordable to most people. **(01 Mark)**
(vii) Should be easy and safe to transport and store. **(01 Mark)**
(viii) Should not have harmful effects on the environment during its production **(01 Mark)**

Any five points (05 marks)

(b) It is not advisable to sleep inside a poorly ventilated house with a burning wooden charcoal because, in a limited supply of oxygen, the burning charcoal produces a carbon monoxide gas which is poisonous. (**01 Mark)**

equation: $2c_{(s)} + O_{2\,(g)} \longrightarrow CO_{(g)}$ **(01 Mark)**

4. (a) (i) Consider the formula of iron(iii) Chloride $FeCl_3$ Let the Oxidation of iron be Fe.

$FeCl_3 = 0$

$Fe + (-1x3) = 0$ ***(02 Marks)***

$Fe - 3 = 0$

$Fe = 0 + 3$

$Fe = +3$

Oxidation number of iron is $^{+}3$ ***(01Mark)***

(ii) A reducing agent is hydrogen **(01Mark)**
An oxidizing agent is copper (ii) oxide **(01Mark)**

(b) (i) Reduction reaction (***00½*** Mark)
(ii) Oxidation reaction (***00½*** Mark)
(iii) Oxidation reaction (***00½ Mark)***
(iv) Reduction reaction (***00½*** Mark)

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5. (a) (i) In pharmacy the knowledge of chemistry is applied in making medicine **.(01Mark)**

(ii) In Agriculture the knowledge of chemistry is applied in making agricultural chemicals like insecticide, pesticides and fertilizer for improving the yield of crops. (0**1mark)**

(b) (i) Water is not used to extinguish fire class B, since the combustible materials is flammable liquid which is less dense than water and hence floats over water and fire will continue burning, but for class E the burning is electrical appliances which has got charges that accelerate burning. (02**Marks)**

(ii) Because dry wooden stick does not conduct electricity. (0**1.5mark)**

(iii) To be easy for people to get out when accident is happened**. (01.5mark)**

6. (a) The modern periodic law states that, "Properties of the elements are periodic function of their atomic numbers". ***(01 Mark)***

(b) (i) - X = 2:8:7 (***00½*** Mark***)***

- Y = 2:8:8 (***00½ Mark)***

- Z = 2:8:8:2 (***00½ Mark)***

(ii) Element Y ***(00½ Mark)***

(iii) ZX_2 ***(01 Mark)***

(iv) Characteristics of a compound formed in (iii)

- Formed between metal and non-metal. **(01 Mark)**
- conduct electricity in molten or solution form. **(01 Mark)**
- Insoluble in organic solvents like benzene. **(01Mark)**

7. (a)

Element	C	H	**O (001/2 Mark)**
% composition	**52.2**	13	**34.8 (001/2 Mark)**
RAM	12	**1**	***16*** **(001/2 Mark)**
Divide % composition by RAM	$\frac{52.2}{12}$ ***= 4.35***	$\frac{13}{1}$ ***= 13***	$\frac{34.8}{16}$ ***=2.18*** **(001/2 Mark)**
Divide by the smallest number	$\frac{4.35}{2.18}$ ***= 1.99∼2***	$\frac{13}{2.18}$ ***= 5.96∼6***	$\frac{2.18}{2.18}$ ***= 1*** **(001/2 Mark)**
The whole number ratio	**2**	***6***	***1***

The empirical formula is C_2H_6O (001/2 Mark)

Find molecular mass

Molecular mass = vapour density × 2 **(001/2 Mark)**

23 × 2 = 46

Therefore molecular mass = 46 **(001/2 Mark)**

Find molecular formula

Molecular mass = (E.F)n

46 = (C_2H_6O)n **(001/2 Mark)**

46 = ((12x2) +(1x6)+(16x1))n

= (24 + 6 + 16)n

46 = 46n

46/46 =n

n= 1 **(001/2 Mark)**

therefore molecular formular =(E.F)n

=(C_2H_6O)n

=(C_2H_6O)1

C_2H_6O = C_2H_6O **(001/2 Mark)**

Therefore the molecular formula of a compound is C_2H_6O

The possible isomers are **CH_3CH_2OH** and **CH_3OCH_3** **(01mark)**

(b) Compound A can decolourise potassium permanganate /bromine water but compound B can not. **(001/2Mark)**

8. (a) Reason for treatment and purification of water.

- To kill disease causing micro-organisms such as bacteria, fungi and amoeba **(01 marks)**
- To remove toxic substance dissolved in water. **(01 marks)**
- To remove suspended earthy materials (clay matter) **(01 marks)**
- To remove odour and unpleasant smells caused by different contaminant dissolved in water **(01 marks)**
- To remove water hardness **(01 marks)**

any five points (05 marks)

(b) (i) Chlorine for killing of microorganisms **(01 marks)**

(ii) Aluminium sulphate allow sedimentation of mud **(01 Marks)**

9. (a) Decrease in temperature: since the reaction is exothermic the decrease in temperature will favor production of ammonia. **(02 Marks)**

(ii) Increase in pressure. Under high pressure the speed of reaction between nitrogen and hydrogen molecules will increase hence produce more ammonia **(02 marks)**

(iii) Concentration: Increase in the concentration of nitrogen and hydrogen will favour forward reaction and hence more ammonia is produced **(02 marks.)**

(b) The equilibrium will be attained when the rate of forward reaction is equal to the rate of backward reaction. **(01 marks.)**

10. (a) (i) because it absorbs water found on the atmosphere and forms hydrated copper (II) Sulphate which is blue in colour. **(01 marks.)**

(ii) Because carbon dioxide is denser than air. **(01 marks.)**

(iii) Because concentrated sulphuric acid reacts with hydrogen sulphide gas.

$H_2SO_{4(l)} + H_2S_{(g)} \longrightarrow 2H_2O_{(l)} + SO_{2(g)} + S_{(s)}$ **(01 marks.)**

(iv) Paraffin floats over the sodium and hence prevent reaction between sodium and other substance like air and water vapour **(01 marks.)**

(b) (i) Ar (ii) Mg (iii) Ag (iv) S (v) O (vi) P **(03 marks @)*00½ Mark)***

11. (a) (i) Non-electrolyte is a solution or molten compound which does not allow passage of electric current through it. ***(00½ mark)***,
Example: wax, sugar, methyl benzene, kerosene. ***(01 Mark @) 00½ mark)***

(ii) Electrolyte is a solution or molten compound which allows passage of electric current through it.
(00½ mark)
Example: Sodium hydroxide, sulphuric acid, sodium chloride, aluminium oxide.
(01 Mark @) 00½mark)

(b) (i) Because when the copper ions from the solution discharge, the copper anode also dissolves in the electrolyte and lose electrons to replace the copper ions in electrolytes hence the nature of the electrolyte remains the same. ***(01 Mark)***
At anode: $Cu_{(s)} \rightarrow Cu^{2+}_{(aq)} + 2e^-$ ***(01 Mark)***
At cathode: $Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$ ***(01 Mark)***

(ii) Because when beryllium is burned in air it forms a very hard protective coat which forbids further reaction. ***(01 Mark)***

12. (a) Data given
Mass of Fe_2O_3 = 80kg = 80,000g
Mr of Fe_2O_3 = 160g
Mr of Fe = 56g **(001/2 Mark)**
The equation
$Fe_2O_{3(S)} + 3CO_{(g)} \longrightarrow 2Fe_{(s)} + 3CO_{2(g)}$ ***(01 Mark)***
From the above equation
160g of Fe_2O_3 = 112g of Fe
(01 Mark)
80,000g of Fe_2O_3 = xg of Fe

$$X = \frac{80{,}000g \times 112g}{160g} \quad \textit{(01 Mark)}$$

= 56,000g = 56kg

The mass of iron produced = 56kg (**001/2 Mark)**

(b) (i) Aluminium is used for making cooking utensils because it is a good conductor of heat.

(01 Mark)

(ii) If aluminium chloride react with water, a white precipitate of aluminium hydroxide is

formed. $AlCl_{3(aq)} + 3H_2O_{(l)} \longrightarrow Al(OH)_{3(s)} + 3HCl_{(aq)}$ ***(01 Mark)***

(iii) Because concentrated nitric acid reacts with aluminium and forms a protective oxide layer.
This protective layer makes aluminium unreactive towards HNO_3 acid.

But aluminium containers can't be used to transport concentrated sulphuric acid because the acid corrodes the container. ***(01 Mark)***

13. Methods used to manage loss of plant nutrients from the soil.

(i) Prevention of soil erosion
(ii) Crop rotation
(iii) Good harvesting practices
(iv) Controlled grazing
(v) Intercropping (Mixed cropping)
(vi) Uses of manures
(vii) Minimum tillage
(viii) Agroforestry
(ix) Mulching

- A candidate should explain any 6 points @ **02 = 12 Marks**
- Introduction **(01$_{1/2}$ Marks)**
- Conclusion **(01$_{1/2}$ Marks)**

14. **Introduction**

Water pollution: This is an addition of harmful material to the water and hence make water unfit for human consumption. **(01$_{1/2}$ mark)**

The following are the causes of water pollution

Industrial waste: When industrial wastes like mercury, Zinc and copper compound are deposited as wastes and dumped in ocean pollute water. **(02 Marks)**

Bad fishing method: This includes use of dynamite and poison in fishing which in turn pollute water and cause death of aquatic organisms. **(02 Marks)**

Agricultural chemicals: When the agricultural chemical like insecticides, pesticides are applied to plants and finally washed by rain into the water sources it will pollute water. **(02 Marks)**

Domestic waste: Human activities like agricultural practices contaminate water with chemicals that pollute water sources. **(02 Marks)**

Radioactive wastes: When radioactive wastes are dumped into the ocean will affect the life of aquatic organisms. **(02 Marks)**

Any three points **(6marks)** @ **2marks)**

The following are the measures to be taken

Recycling of industrial waste: In order to avoid water from being polluted the industrial wastes should be recycled. **(02 Marks)**

Use of manure and reliable ways of managing land: The farmers should be encouraged to use manure instead of artificial fertilizer and reliable ways of managing land (**02 Marks)**

Educating societies: A proper education concerning ways of sewage disposal should be provided to the societies. **(02 Marks)**

Laws against sewage disposal: The strict law against the disposal of the harmful wastes from industries and domestic should be in acted. **(02 Marks)**

Any three points **(06 marks)** @ **02 marks)**

Conclusion

More effort should be done by both government and society in general on the ways of avoiding water pollution in order to safe with the effects of water pollution. **(01 1/2 mark)**